

Rudraksh Mohapatra

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EDUCATION

Stanford University

GPA: 3.8

M.S Electrical Engineering (Coterminal Program)

June 2028 (expected)

B.S Electrical Engineering

June 2027 (expected)

Fellowships: Rise Global Fellow (Rhodes Trust and Schmidt Futures, Top 100 selected from 80,000+ applicants globally)

Coursework: 3D Imaging Sensors, Signals, Circuits, Digital Systems Design, Probability, Fourier Methods, Differential Equations & Linear Algebra, Computer Organization and Systems

EXPERIENCE

MorphoAI - Software Engineering Intern

New York, NY

YC Backed Startup building generative design software for robotics and machines

Jun. 2026 - Sep. 2026

- Built a constrained nonlinear optimizer that deforms 3D solid models to hit exact volume, surface-area, and per-feature dimensional targets while minimizing vertex displacement. Optimizer solved a joint decision vector of mesh vertices and analytic surface parameters with a custom sparse Gauss–Newton KKT solver.
- Derived analytic residuals and Jacobians for primitive shapes and NURBS surfaces. Designed membership constraints, holding every deformed vertex exactly on its parametric surface.
- Shipped STEP round-trip export and demoed to clients. Work was selected for continued development.

RESEARCH & PROJECTS

Navigation and Autonomous Vehicles Lab

Stanford, CA

Gaussian Splatting for 3D Scene Reconstruction and Analysis

Sep. 2025 - Present

- Analyzed spherical-harmonic coefficients to characterize geometry/view-dependent-color coupling.
- Implemented density-based clustering algorithms (DBSCAN, HDBSCAN, OPTICS) to segment 3D scenes and improve artifact detection in low-texture regions.
- Presented work at Stanford Sustainability and Data Science Conference 2026 and American Geophysical Union 2026.

Stanford Smart Sensing Systems Lab

Stanford, CA

Algorithm Development for SAR-Based Satellite Backscatter Communication

Jun 2025 - Present

- Implemented a signal processing algorithm for passive RF backscatter communication using existing SAR infrastructure. Partitioned the aperture into sublooks to encode bits, performed range/azimuth compression, and computed Signal-to-Clutter Ratio (SCR) per sublook and across sublooks.
- Wrote a Radar Cross Section (RCS) estimation program and validated with theoretical estimates to within 2 dB.
- Simulated corner reflector signatures in raw SAR data by modelling a point target response in the collected I/Q samples, generating controlled ground-truth test cases for backscatter experiments without field deployment.

Projects

Stanford, CA

- **Programmable Image Projection Using Custom Diffraction Masks (2026 Design Award - Stanford EE Department)**
Designed an optical system that recreates images from their custom diffraction masks. Diffraction patterns were generated using diffractsim python libraries.
- **Face Liveness Detection System using 3D Time of Flight Camera:** Built a liveness detection system using 3D Time of Flight Camera to distinguish real faces from 2D spoofs. Utilized near/far mode and FoV of the camera to obtain depth information and histogram data to gather insights.
- **Sensing Systems (EE117: Top 2 of 24 teams):**
IMU Dead Reckoning: Built a gravity level measurement using Accelerometer and Gyroscope registers (no fusion libraries) and applied a Kalman filter for position tracking of target.
Capacitive Touch Sensor: Localized finger tracking at 15 fps by detecting the amount of pressure based on autocorrelation values of PRBS sequence readings from the drive lines.

SKILLS & CERTIFICATIONS

- **Programming Languages and Workflows:** Python, C, C++, Verilog, Bash, Git, MATLAB, FORTRAN
- **Libraries:** NumPy, SciPy, OpenCASCADE, Cyipopt
- **Hardware:** Raspberry Pi, FPGA (PYNQ), optical bench/laser alignment, LTSpice, Oscilloscope, VNAs